Jingyun Yang

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Education

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Machine Learning · GPA: 4.12/4.33 — *August 2020 – December 2021*

University of Southern California — Los Angeles, CA
Bachelor of Science in Computer Science · Minor in Mathematics · GPA: 4.00/4.00 — *August 2016 – May 2020*

Publications

1. **Jingyun Yang**, Junwu Zhang, Connor Settle, Akshara Rai, Rika Antonova, Jeannette Bohg, "Learning Periodic Tasks from Human Demonstrations", in submission.
2. Rika Antonova, **Jingyun Yang**, Priya Sundaresan, Dieter Fox, Fabio Ramos, Jeannette Bohg, "A Bayesian Treatment of Real-to-sim for Deformable Object Manipulation", in submission.
3. **Jingyun Yang***, Hsiao-Yu Fish Tung*, Yunchu Zhang*, Gaurav Pathak, Ashwini Pokle, Christopher G. Atkeson, Katerina Fragkiadaki, "Visually-Grounded Library of Behaviors for Manipulating Diverse Objects across Diverse Configurations and Views", Conference on Robot Learning (CoRL), 2021.
4. Youngwoon Lee, **Jingyun Yang**, Joseph J. Lim, "Learning to Coordinate Manipulation Skills via Skill Behavior Diversification", International Conference on Learning Representations (ICLR), 2020.
5. Karl Pertsch, Oleh Rybkin, **Jingyun Yang**, Konstantinos Derpanis, Kostas Daniilidis, Joseph J. Lim, Andrew Jaegle, "KeyIn: Keyframing for Visual Planning", Learning for Dynamics and Control (L4DC), 2020.

Experience

CMU Professor Katerina Fragkiadaki's Lab — Pittsburgh, PA
Research Assistant · Supervisor: Katerina Fragkiadaki — *June 2020 – Present*

- **Efficient Exploration with Kinematics Prediction:** Investigating exploration methods that predict and finetune the kinematic structure of a scene while interacting with it using the predicted scene structure.
- **Visual Representations for Generalizable Manipulation:** Designed and implemented a library-based framework that learns visual representations to help robotic agents generalize to varying objects, initial configurations, and camera viewpoints. Paper accepted in CoRL 2021.

Stanford Interactive Perception and Robot Learning (IPRL) Lab — (Remote) Pittsburgh, PA
Summer Intern · Supervisor: Jeannette Bohg — *June 2021 – September 2021*

- **Policy Representations for Periodic Tasks:** Explored appropriate representations for periodic tasks such as table wiping and rope winding.
- **Efficient Learning of Manipulation Skills from Human Demonstrations:** Developed an approach that can efficiently learn to manipulate granular and deformable objects from human demonstrations.

USC Cognitive Learning for Vision and Robotics (CLVR) Lab — Los Angeles, CA
Research Assistant · Supervisor: Joseph J. Lim — *August 2018 – May 2020*

- **Learning Coordinated Tasks from Rewards:** Designed, implemented, and tested a hierarchical reinforcement learning model that tackles complex tasks of multiple agents by learning composable primitive skills and coordination of the skills. Paper accepted in ICLR 2020.
- **Video Prediction with Temporal Hierarchy:** Developed experiments and designed downstream planning algorithm for a model that learns to discover informative frames in a future video sequence and represent the video via its keyframes. Paper accepted in L4DC 2020.

Bloomberg L.P. — New York, NY
Software Engineering Intern · Supervisor: Luke Shen — *May 2018 – August 2018*

- **Backend Software Development:** Migrated core functionalities of the Company Financials Core Algorithms Library to Spark in Scala.

Awards and Honors

- USC University Trustees Award — *April 2020*
- USC Provost's Undergraduate Research Fellowship — *May, August 2019*
- Tau Beta Pi the Engineering Honor Society — *April 2018*